EVAN J. LONG

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EDUCATION

The University of Texas at Austin – Austin, TX -- *Aug. 2023 – May 2027*
Bachelor of Science in Mechanical Engineering | GPA: 3.71/4.00 | 112 Course Hours
Relevant Coursework: Machine Elements, Material Analysis, Fluids / Heat Transfer, Robotics, Dynamics Systems and Controls, Coding.
The University of New Mexico - Los Alamos – Los Alamos, NM --- *May 2021 – May 2023*
Certificate in Robotics | GPA: 3.86/4.00 | 33 Course Hours
Relevant Coursework: Advanced/Industrial Robotics Projects; Manufacturing Techniques; Hydraulic, Cryogenic, and Vacuum systems.

EXPERIENCE

ADAPT Prosthetics for Animals – *Design team / Production (Machining) and FEA Specialist* ----------------------------- *Jan. 2026 – Present*
- Taught members how to conduct effective analyses of their parts, reducing material use and increasing overall strength.
- Conducted photogrammetry on a dog's leg to create a snug, durable, comfortable prosthetic to ease her pain while walking.

Hello Maker Studio – *Maker / Mentor / Researcher* --- *Aug 2024 – Present*
- Designed and produced various low-cost scientific apparatuses for CNS research labs, streamlining their process workflows
- Developed comprehensive machine training modules, and implemented a feedback loop to improve user experience.
- Mentor students and faculty across colleges and disciplines to refine their designs and manufacturing techniques.

ASCEND Autonomous Paraglider – *Chief Mechanical Engineer* -- *Jan. 2024 – Present*
- Engineered and produced multiple robust assemblies to autonomously steer the ground vehicle (tube-frame ATV).
- Designed and produced low-cost test stands to verify thrust specs and fine-tune the motor and propeller arrangement.
- As Production Lead delegated tasks and taught manufacturing to a team of 4. Responsible for all mechanical components.

EXD Experimental Division Combat Robotics – *Botfighter* -- *Sep. 2024 – Present*
- Designed 5 different PLAntweight battlebots. Fought at multiple intercollegiate competitions, recently placing 2nd overall.
- Optimized material selection, geometry, manufacturing and assembly techniques to create a robust competitive 3lb bot.

Texas Rocket Engineering Lab (TREL) – *Ground Systems – Vehicle/Stage Test Responsible Engineer* ----------------- *Jan. 2024 – Aug. 2025*
- Quadrupled hardware output over previous semesters despite team size halving through leading the ground fluids team in developing and building systems to store: gaseous hydrogen, gaseous and liquid oxygen, helium, nitrogen, RP-1, and water.
- Enhanced project planning, documentation, and parts sourcing; fostering collaboration with electronics hardware, software, and structures teams to establish reliable testing protocols and autosequencing while managing P&ID creation and analysis.
- In overseeing the design, testing, and integration of fluid systems, prioritized safety, effectively managing risks associated with dangerous cryogens and explosive oxidizers and fuels; developed nearly the entirety of the FireX system by optimizing layout and pressure specifications to effectively protect the engine bay, seamlessly integrating with other test hardware.
- Pioneered a mobile GSE trailer, redesigning and reassembling different transfer configurations from changing requirements.

Systems for Augmenting Human Mechanics (SAHM) Lab – *Undergraduate Researcher, General Engineer -------- Oct. 2023 – Jan. 2025*
- Analyzed research documents and produced an orthosis to assist patients with impaired mobility in walking, by using bands to redistribute work done by rear leg muscles to assist the degenerated ones in the front, easing the overall swing phase.
- Set up and applied an electromyography (EMG) system to analyze neuromuscular signals, displaying raw signal strength and converting muscle activity into actionable data to control simulated or real prosthetics.

Nuclear and Applied Robotics Group (NRG) – *Undergraduate Researcher (FIRE)* --------------------------------------- *Jan. 2024 – May 2024*
- In a team of 3, developed a cutting-edge 6DOF Ceramic 3D Printer module, adaptable to any URD3 robotic arm.
- Owned the entire extruder assembly, prototyped complex housing and augur geometries to evenly lay high-viscosity clay.

Relevant Course Projects, UNM – *Student* --- *Jan. 2022 – May 2023*
- Designed, built, and wired a functional robotic tank featuring a high-powered 2-axis pneumatic cannon. Its abilities were put on display by the university. I was appointed ambassador to local schools, encouraging students to join a robotics program.
- Developed end effectors for and programmed a FANUC standard industrial robot, enabling it to make, pour, and serve tea

COMMUNITY INVOLVEMENT

Helping Hand Home – Austin, TX -- *Jan. 2025 – Present*
- Created, tested, sourced parts for, and performed fun interactive Science Demos to nurture students interest in STEM fields.
- Forged valuable personal relationships with children (age 7-12) who require more attention that foster care can provide.

RELEVANT SKILLS

- **Design Software**: SolidWorks (including FEA), OnShape, SketchUp, Fusion360 (CAM), MakeraCAM, FDM & SLA 3D Printing
- **Robotic Systems and Programming**: Python, C++, ROS, General Electronics Layout and Assembly.
- **Technical Certifications**: FANUC Industrial Robot Technician I and II, Programmable Logic Controllers, Safety Certifications (Fire, Cryogen, Compressed Gases, General Laboratory), Numerous machining ones, including manual mill, lathe, CNC mill